CAS4133 LLM · 기말 대비 · 기출 & 출제 예측

기출문제 모음 & 기말 출제 예측

유일한 기출인 2025 봄 중간고사 전체 원문(30문항) + 시험 형식·전략 분석 + 기말 범위(#7~#14) 출제 예측 — 마지막 수업 공지(객관식 3~4개·특강 통합 Problem 확정) 반영

📅 기출: 2025-04-22 중간고사 (Jaehyung Kim) | 📊 100점 / 100분 | ⏱ T/F 10 + Pick-all 20 = 30문항

📑 목차

1 시험 형식 분석 & 풀이 전략

1.1 시험 구조 (2025 중간 기준)

Problem	형식	문항 수	배점	채점 규칙
Problem 1	True / False	10문항	30점 (각 3점)	정답 **3점** / 오답 **0점** / **무응답 1점**
Problem 2	Pick all that are correct	5문항	15점 (각 3점)	"any wrong answer will lead to **0 points**" — 보기 (a)~(d) 중 정답 조합을 **완벽히** 골라야 득점. 부분점수 없음
Problem 3	Pick all that are correct	5문항	20점 (각 4점)	
Problem 4	Pick all that are correct	5문항	20점 (각 4점)	
Problem 5	Pick all that are correct	5문항	15점 (각 3점)	
합계		30 문항	**100점**	시험시간 100분 (문항당 평균 3분 이상 — 시간은 충분)

📢 **마지막 수업 공지 반영** 확정

- **객관식(Pick-all/multiple choice) Problem은 3~4개** — 교수 직접 발언: *"I will make three or four multiple choice question problems"*. 확정
- **그중 1개 = 특강 통합 Problem**: *"one of them will be about the jailbreaking and AI-generated detection, all in one part"* — #13 Jailbreak + #14 AI Text Detection이 **한 Problem에 혼합 출제**. 확정
- 중간 패턴상 **T/F Problem(10문항)은 별도로 있을 가능성이 큼** — 교수는 객관식 개수만 언급했으므로 이는 추정. 추정

- **성적**: 중간 20% / **기말 30%** / 과제 20% / 프로젝트 20% / 출석 10%. **절대평가** — A·A-는 후하게, A+는 엄격하게. 확정

⭐ **구조에서 읽히는 출제 패턴**

- **챕터당 1개의 Problem**(5문항 내외) + T/F는 전 범위에서 골고루 10문항. 기말도 같은 틀일 가능성이 높다 — 기말 범위 8개 챕터 중 **4개 내외가 Problem으로 승격**되고, 나머지는 T/F에서 소화될 것.
- Pick-all 보기 4개 중 정답 수는 1~4개 모두 등장 (기출: 정답 1개 ×2, 2개 ×9, 3개 ×8, 4개 ×1). **"정답이 보통 2개"라는 가정은 위험** — 전부 정답(P3-1)이나 단 하나(P3-2, P4-4)도 나온다.
- 해설 문체가 정형화: "Correct. ..." / "Incorrect. ..." — 각 보기가 독립적인 T/F 명제다. 즉 **Pick-all = 미니 T/F 4연발**로 풀면 된다.

1.2 점수 전략

1. **T/F 무응답 전략 (기대값 계산)** — 무응답은 확정 1점. 찍기의 기대값은 $0.5 \times 3 = 1.5$점이므로 **50:50이라도 찍는 게 기대값상 유리**하다. 단, "확신이 40% 미만"(상대가 더 그럴듯해 보일 때)이면 기대값이 1.2점 이하로 떨어져 무응답(확정 1점)과 차이가 미미 — 이때는 **리스크 없는 1점 확보**가 합리적. 결론: *완전히 모르는 문제만 비우고, 조금이라도 기울면 답하라.*
2. **Pick-all은 보기별 독립 판정** — 각 보기를 True/False로 따로 채점한 뒤 True만 모은다. "이 중 2개가 답이겠지" 식의 조합 추론 금지. 한 보기라도 틀리면 0점이므로, **확신 없는 보기는 '강한 근거가 없으면 제외'**가 아니라, **해당 보기의 명제가 참인지 자체를 판정**해야 한다. 애매한 보기 처리: ①절대어(always/only/never/all)가 들어 있으면 False 쪽으로 기울인다(아래 함정 유형 ②). ②수치·고유명사는 강의 슬라이드의 정확한 값과 대조한다. ③"일반적으로 맞지만 예외가 있는" 서술은 출제자가 Correct 처리하는 경향(예: P2-1(a) data sparsity).
3. **시간 배분** — 100분/30문항. 1회독(60분) → 애매 문항 재검토(25분) → T/F 무응답 최종 결정(5분) → 마킹 검산(10분). 50분 후 퇴실 가능하지만 만점이 목표면 끝까지 검토.

1.3 함정 패턴 유형화 — 기출 해설의 "Incorrect" 전수 분석

기출 30문항의 모든 오답 보기("Incorrect. ...")를 유형별로 분류하면 출제자의 함정 제조법이 보인다. **기말에도 같은 틀로 함정이 만들어진다.**

유형	함정 제조법	기출 실례	기말 예상 적용
① 주체 바꿔치기 swap	A의 속성을 B에 붙인다 (encoder↔decoder, draft↔target model)	P2-4(d) "The *encoder* uses causal attention" → decoder의 속성. P2-5(c) "only applicable for encoder" → decoder에도 적용	retriever↔reader(RAG), actor↔critic, attacker↔defender(jailbreak), generator↔verifier
② 절대어 함정 always/only/all	대체로 맞는 명제에 always·only·all·no·eliminates를 박아 False로 만든다	P1(x) DBS "*always* chooses the most probable", P2-3(b) "RNNs *always* preserve long-term dependencies", P3-3(b) "MoE activates *all* experts", P5-4(c) "*eliminates* the need for sequential generation"	"RAG always improves factuality", "Tool use eliminates hallucination", "RL requires no reward model" 류
③ 수치 함정 number	유명한 수치를 다른 모델의 수치로 치환	P3-2(c) "GPT-3 scales up to *110 million*" → 175B (110M은 BERT-base)	파라미터 수, top-k의 k, 벤치마크 수치, 논문 연도

④ 방법 귀속 오류 attribution	방법 X의 주장/기법을 방법 Y의 것으로 서술	P3-4(a) "model size should be prioritized" → Kaplan의 주장 (Chinchilla 아님). P4-2(d) "supervised loss on preference pairs" → DPO의 방법(RLHF 아님). P4-4(a) "DPO requires a separate reward model" → RLHF의 특징	DPR↔BM25, Toolformer↔ToolLLM, ReAct↔Reflexion, PPO↔GRPO, GCG↔PAIR, DetectGPT↔watermarking
⑤ 처리 순서/위치 함정 order	연산이 적용되는 단계를 뒤바꾼다	P5-2(d) "Temperature is applied *after* the softmax" → logits에 적용 (before softmax)	"retrieval happens after generation", "KL penalty applied after reward" 류
⑥ 범주 혼동 category	inference 기법↔training 기법, online↔offline, discrete↔continuous를 섞는다	P5-5(b) test-time compute = "fine-tuning technique" → inference 전략. P4-4(d) "DPO is inherently *online*" → offline. P2-2(a) tokenization → "*continuous*	"RAG는 학습 기법", "speculative decoding은 학습 가속", "DPO·GRPO online/offline" 구분

		units" → discrete. P5-5(d) "increasing the number of *parameters* at test time" → 파라미터는 고정	
⑦ 목적/동기 왜곡 motivation	방법의 진짜 동기를 그럴듯한 다른 동기로 바꾼다	P1(viii) preference learning의 main motivation = "held-out 성능 향상" → alignment가 본 동기. P4-3(c) DPO가 "minimize the difference" → 상대 log-likelihood를 maximize	"RAG의 동기는 속도", "watermark의 동기는 품질 향상" 류

⚠ **실전 체크리스트 (보기 읽을 때마다)**

- 절대어(always / only / all / never / eliminates / requires / cannot / any)가 보이면 → 반례 1개를 떠올려본다. 반례가 있으면 False.
- 주어(모델/방법 이름)와 술어(속성)가 정말 그 조합이 맞는지 — "이 속성은 원래 누구 것?"을 자문한다.
- 수치는 외운 값과 정확히 대조 (175B, 110M, top-k의 k, Chinchilla 20 tokens/param 등).
- 이 기법이 **training-time인지 inference-time인지**, online인지 offline인지 분류부터 한다.

- 단, 함정 같아 보여도 전부 맞는 문제(P3-1: 정답 a,b,c,d)도 있다 — 패턴 매칭이 아니라 명제 자체를 판정하라.

2 Problem 1: True / False (30 Points) 2025 중간 기출 원문

"Identify whether the following statements are True or False. You will get 3 points for each correct answer, 0 points for each wrong answer, and 1 point for each unanswered question."

(i) True/False · 3 pts 2025 중간 기출 중간 범위 (N-gram)

A bigram model assumes that each token depends on all previous tokens in the sequence.

▼ 정답 & 해설 보기

False. A bigram model assume that each word only depends on the last previous word in the sentence. ② 절대어/정의 왜곡

(ii) True/False · 3 pts 2025 중간 기출 중간 범위 (Tokenization)

Vocabulary size is a hyperparameter that can be defined by the user in Byte-Pair Encoding.

▼ 정답 & 해설 보기

True. Unlike character-level or word-level tokenizers, BPE can control a vocabulary size.

(iii) True/False · 3 pts 2025 중간 기출 중간 범위 (RNN/LSTM)

Seq2seq with LSTM can theoretically handle sequences of any length.

▼ 정답 & 해설 보기

True. There is no limitation in theory while it suffers issues such as GPU memory or limited performance from forgetting issues in practice. *주의: "theoretically"가 들어가면 True일 수 있음 — 절대어 함정의 역방향.*

(iv) True/False · 3 pts

2025 중간 기출 | 중간 범위 (Transformer)

Positional encoding is learned via backpropagation in standard Transformers.

▼ 정답 & 해설 보기

False. As we leanred in lecture, the original Transformer paper introduces the fixed rule utilizing sign and cosine functions to generated positional encoding.

⑥ 범주 혼동 (learned⇔fixed)

(v) True/False · 3 pts

2025 중간 기출 | 중간 범위 (Pre-training data)

Model-based data filtering refers to a technique that uses LLM's in-context learning to filter out all the low-quality data.

▼ 정답 & 해설 보기

False. As size of pre-training data is too large to apply LLM for all data, it is applied for limited number of data to generate labeled data and the separate classifier is trained on that.

② 절대어 ("all the low-quality data")

(vi) True/False · 3 pts

2025 중간 기출 | 중간 범위 (Emergent abilities)

In-context learning is an example of an emergent ability in LLMs.

▼ 정답 & 해설 보기

True. Small LLMs do not exhibit in-context learning capability and it is not predictable with scaling law.

(vii) True/False · 3 pts

2025 중간 기출 | 중간 범위 (Instruction tuning)

FLAN fine-tunes models to follow instructions using cross-entropy loss.

▼ 정답 & 해설 보기

True. (해설 원문 없음 — FLAN은 instruction 데이터에 대한 supervised fine-tuning, 즉 표준 cross-entropy loss 사용.)

(viii) True/False · 3 pts
2025 중간 기출 기말 연관: #8 AI Feedback (alignment 동기)

Main motivation of preference learning is improving model performance on held-out datasets.

▼ 정답 & 해설 보기

False. Main motivation is improving alignment; it can compete with model performance like accuracy on held-out test datasets. ⑦ 목적/동기 왜곡

(ix) True/False · 3 pts
2025 중간 기출 기말 연관: #12 RL (preference optimization 계열)

SimPO introduces a length-normalized reward and no reference model.

▼ 정답 & 해설 보기

True. Unlike DPO, SimPO introduces length-normalization term in implicit reward, and removes the reference model as well.

(x) True/False · 3 pts
2025 중간 기출 기말 연관: #7 Decoding (DBS)

Diverse Beam Search (DBS) always chooses the most probable sequence (i.e., the lowest perplexity) from all beams.

▼ 정답 & 해설 보기

False. As the diversity term is introduced, the selected sequence can be not optimal in the perspective of perplexity. ② 절대어 ("always")

⭐ **Problem 1에서 배울 것**

T/F 10문항 중 **기말 범위와 직결된 것이 3개**((viii)(ix)(x)) — 기말 T/F도 직전 범위(중간 이후 #7~#14)에서 전 챕터 골고루 1~2문항씩 출제될 것. False 5개 중 4개가 절대어·동기왜곡·정의왜곡 패

턴.

3 Problem 2: N-gram, RNN, Transformer (15 Points)

2025 중간 기출 원문

"Answer the following questions about N-gram, RNN, Transformer." — 모든 문항: Pick all that are correct; any wrong answer will lead to 0 points.

중간 범위 기말에 직접 재출제될 가능성은 낮지만, **Pick-all 문항 설계 방식(보기 4개 = 독립 T/F 4개)**의 표본으로 정독할 가치가 있다.

2-1. Pick all that are correct · 3 pts 2025 중간 기출

What are the typical limitations of N-gram language models?

a. They suffer from data sparsity, especially for large N.

b. They can model arbitrarily long context sequences effectively.

c. They struggle with Out-of-Vocabulary (OOV) words.

d. The input space size grows exponentially with the value of N.

▼ 정답 & 해설 보기

정답: (a), (c), (d)

- (a) Correct. Data sparsity is a major issue—most word sequences don't appear often in the corpus.
- (b) Incorrect. N-gram models can only handle a fixed-length window and do not capture long contexts.
- (c) Correct. OOV words are problematic since they are not in the vocabulary and hence not modeled.
- (d) Correct. The number of possible N-grams grows exponentially with N and vocabulary size.

2-2. Pick all that are correct · 3 pts 2025 중간 기출

Which of the following statements about tokenization in language models is true?

a. Tokenization is the process of converting input text into a sequence of continuous units.
b. Subword tokenization typically results in smaller vocabulary compared to word-level tokenization.
c. Byte-Pair Encoding is not used in any state-of-the-art language models.
d. Character-level tokenization leads to longer input sequences compared to word-level tokenization.

▼ 정답 & 해설 보기

정답: (b), (d)

- (a) Incorrect. Tokenization maps text into a sequence of tokens (i.e., discrete units), which are model input units. **⑥ continuous⇔discrete**
- (b) Correct. Subword-level tokenization typically reduces vocabulary size by breaking words into smaller units.
- (c) Incorrect. BPE is used in many modern LMs such as GPT and Roberta. **② 절대어 ("not used in any")**
- (d) Correct. Character-level tokenization results in longer sequences due to finer granularity of vocabulary.

2-3. Pick all that are correct · 3 pts 2025 중간 기출

Which of the following are correct about the limitations of vanilla RNNs?

a. RNNs are prone to vanishing gradients during backpropagation through input time steps.
b. RNNs always preserve long-term dependencies thanks to their recursive structure.
c. Exploding gradients can occur in RNNs and can be mitigated using gradient clipping.
d. Vanilla RNNs require a fixed sequence length to function properly.

▼ 정답 & 해설 보기

정답: (a), (c)

- (a) Correct. Vanishing gradients are a major issue in vanilla RNNs when modeling long sequences.
- (b) Incorrect. RNNs often fail to preserve long-term dependencies due to vanishing gradients. ② 절대어 ("always")
- (c) Correct. Exploding gradients can occur and are commonly managed using gradient clipping.
- (d) Incorrect. RNNs do not require fixed sequence length; they are flexible with variable-length input. ② 절대어 ("require")

2-4. Pick all that are correct · 3 pts 2025 중간 기출

Which of the following are true about self-attention in Transformers?

a. Self-attention computes interactions among all tokens in the same sequence.

b. The query, key, and value matrices are learned during training.

c. The decoder attends to encoder outputs through cross-attention.

d. The encoder uses causal (masked) self-attention to prevent attending to future positions.

▼ 정답 & 해설 보기

정답: (a), (b), (c)

- (a) Correct. Self-attention models relationships among all tokens in the input.
- (b) Correct. Query, key, and value matrices are trainable and learned during training.
- (c) Correct. Cross-attention is the same attention mechanism introduced in seq2seq model.
- (d) Incorrect. The decoder uses causal attention, not the encoder. ① 주체 바꿔치기 (encoder↔decoder)

2-5. Pick all that are correct · 3 pts 2025 중간 기출

Which of the following are correct regarding multi-head attention in the Transformer?

a. Each attention head has its own set of query, key, and value weights.

b. Multi-head attention forces the model to capture different types of relationships in parallel.

c. It is only applicable for a bi-directional self-attention in the encoder.

d. Using multiple heads increases model capacity without significantly increasing the overall computation cost compared to a single head with larger hidden dimension.

▼ 정답 & 해설 보기

정답: (a), (d)

- (a) Correct. Each head has separate Q/K/V projections to capture different patterns.
- (b) Incorrect. They are not forced to learn different relationship, but has an implicit tendency. **② 강제성 과장 ("forces")**
- (c) Incorrect. It can be also applicable to the decoder or encoder-decoder models. **①+② 주체 한정 ("only ... encoder")**
- (d) Correct. Overall computation is almost same when the dimension of projection layer is divided into the number of multiple heads.

4 Problem 3: Large Language Models (20 Points)

2025 중간 기출 원문

"Answer the following questions about LMs, LLMs, scaling laws, and emergent abilities." — 모든 문항: Pick all that are correct; any wrong answer will lead to 0 points.

중간 범위 단, 수치 함정(GPT-3 175B)·방법 귀속 함정(Kaplan↔Chinchilla)의 교과서적 표본 — 기말에서도 같은 제조법이 반복된다.

3-1. Pick all that are correct · 4 pts

2025 중간 기출

Which of the following correctly compare BERT and GPT?

a. BERT is based on Transformer encoder; GPT is based on Transformer decoder.
b. BERT uses bidirectional attention; GPT uses unidirectional (causal) attention.
c. GPT is better suited for generation tasks like story writing.
d. GPT is pre-trained on next-token prediction, while BERT is trained with masked token prediction.

▼ 정답 & 해설 보기

정답: (a), (b), (c), (d) — 전부 정답!

- (a) Correct. BERT = encoder-only; GPT = decoder-only.
- (b) Correct. BERT allows bidirectional context; GPT is strictly left-to-right.
- (c) Correct. GPT is better suited for generation tasks; BERT is for classification/QA.
- (d) Correct. BERT uses masked LM; GPT uses next-token prediction.

교훈: 4개 전부 정답인 문제도 출제된다. "하나는 틀렸겠지" 가정 금지.

3-2. Pick all that are correct · 4 pts

2025 중간 기출

Which of the following are true about GPT-3's architecture and training?

a. GPT-3 is trained using masked language modeling.

b. GPT-3 introduces in-context learning (ICL) where no parameter updates are needed to learn from examples directly within the input prompt.

c. GPT-3 scales up the number of parameters to 110 million.

d. GPT-3 uses both encoder and decoder modules from the Transformer architecture.

▼ 정답 & 해설 보기

정답: (b) — 단독 정답

- (a) Incorrect. GPT-3 uses next-token prediction, not masked LM. **④ 방법 귀속 (BERT의 방식)**
- (b) Correct. In-context learning is a key feature of GPT-3.
- (c) Incorrect. The largest GPT-3 has 175B parameters. **③ 수치 함정 (110M = BERT-base)**
- (d) Incorrect. GPT-3 uses only the decoder portion of the Transformer. **① 주체 바꿔치기**

3-3. Pick all that are correct · 4 pts 2025 중간 기출

Which of the followings are correctly describe innovations in DeepSeek-V3?

a. Multi-Head Latent Attention (MLA) is proposed to reduce the computational cost such as KV cache.

b. Mixture-of-Experts (MoE) activates all experts during inference.

c. Fill-in-the-Middle (FIM) pre-training requires to modify the architecture as it relies on right-to-left modeling.

d. Multi-Token Prediction (MTP) requires additional training modules to predict multiple tokens at once.

▼ 정답 & 해설 보기

정답: (a), (d)

- (a) Correct. MLA compresses Q/K/V to reduce memory cost.
- (b) Incorrect. MoE activates top-K experts, not all. **② 절대어 ("all experts")**
- (c) Incorrect. FIM does not modify the architecture; instead, it modifies the format of training data. **⑥ 범주 혼동 (architecture↔data format)**

- (d) Correct. MTP requires additional training modules and they can be removed or recycled at inference time.

3-4. Pick all that are correct · 4 pts 2025 중간 기출

Which of the following correctly describe the contributions of the Chinchilla paper?

a. Chinchilla shows that model size should be prioritized over data size under fixed FLOPs.

b. Chinchilla proposes a revised scaling law showing that both model size and data are equally important.

c. Chinchilla achieved better performance than Gopher using the same compute budget.

d. The empirical result supports training smaller models with fewer tokens for optimal performance.

▼ 정답 & 해설 보기

정답: (b), (c)

- (a) Incorrect. This is Kaplan's suggestion; Chinchilla shows both size and data matter equally. **④ 방법 귀속 (Kaplan↔Chinchilla)**
- (b) Correct. Chinchilla revises Kaplan's law to emphasize balance.
- (c) Correct. Chinchilla outperformed Gopher with the same training FLOPs.
- (d) Incorrect. Chinchilla actually promotes smaller models trained on **more** data for optimal scaling. **반쪽 진실 (smaller O / fewer tokens ×)**

3-5. Pick all that are correct · 4 pts 2025 중간 기출

기말 연관: #7 (test-time reasoning과 연결)

Which of the following correctly describe Chain-of-Thought (CoT) prompting in LLMs?

a. CoT requires structured step-by-step reasoning examples during training.

b. CoT enables reasoning capabilities in large language models.

c. The benefit of CoT increases as model size increases.

d. CoT cannot be used in zero-shot settings.

▼ 정답 & 해설 보기

정답: (b), (c)

- (a) Incorrect. While few-shot CoT uses examples, zero-shot CoT does not need them. **②+⑥ ("requires" + training↔prompting)**
- (b) Correct. CoT makes LLMs better at reasoning-intensive tasks like math QA.
- (c) Correct. The gain from CoT is more prominent in larger models.
- (d) Incorrect. Zero-shot CoT (e.g., with triggers like "Let's think step by step") is effective. **② 절대어 ("cannot")**

5 Problem 4: Alignment of LLMs (20 Points)

2025 중간 기출 원문

"Answer the following questions about alignment of LLMs." — 모든 문항: Pick all that are correct; any wrong answer will lead to 0 points.

기말 연관 높음 RLHF·DPO·평가 bias는 기말 범위 **#8 AI Feedback**(RLAIF·LLM-as-a-Judge·LC win rate)과 **#12 RL with LLMs**(PPO·GRPO·DPO 계열)의 직접적 선수 지식이다.

4-1. Pick all that are correct · 4 pts

2025 중간 기출 · 중간 범위 (Instruction tuning)

Which of the following are true about instruction tuning in LLMs?

a. Instruction tuning trains models to follow natural language instructions across multiple tasks.

b. Instruction tuning is only effective for tasks with a single correct answer.

c. Instruction tuning enables zero-shot generalization to unseen tasks.

d. Instruction tuning typically involves supervised fine-tuning (SFT) on labeled instruction datasets.

▼ 정답 & 해설 보기

정답: (a), (c), (d)

- (a) Correct. Instruction tuning uses language-based instructions to guide models.
- (b) Incorrect. It is effective for both objective and subjective tasks. ② 절대어 ("only")
- (c) Correct. It improves zero-shot generalization to new tasks.
- (d) Correct. SFT is commonly used during instruction tuning with instruction-output pairs.

4-2. Pick all that are correct · 4 pts

2025 중간 기출 · 기말 연관: #12 RL (RLHF 파이프라인)

Which of the following are part of the RLHF pipeline used in InstructGPT?

a. Supervised fine-tuning (SFT) on human demonstrations.

b. Training a reward model on human preference data.

c. Fine-tuning the language model using reinforcement learning such as PPO.

d. Directly optimizing language model using supervised loss on preference pairs.

▼ 정답 & 해설 보기

정답: (a), (b), (c)

- (a) Correct. First step of RLHF is conducting SFT on prompts and human-written completions.
- (b) Correct. Reward model is trained using pairwise comparisons.
- (c) Correct. Proximal Policy Optimization (PPO) is used for policy fine-tuning.
- (d) Incorrect. This is DPO's method, not RLHF. **④ 방법 귀속 (DPO↔RLHF)**

4-3. Pick all that are correct · 4 pts 2025 중간 기출 기말 연관: #12 RL (DPO)

Which of the following are correct about DPO?

a. DPO is derived from the same objective as RLHF's PPO fine-tuning.

b. The Bradley-Terry model is used to relate preference and reward.

c. DPO updates language model to minimize the difference between chosen and rejected outputs.

d. DPO approximates reinforcement learning using multi-agent simulation.

▼ 정답 & 해설 보기

정답: (a), (b)

- (a) Correct. DPO reformulates RLHF's objective analytically.
- (b) Correct. It uses the Bradley-Terry model to interpret preference scores.
- (c) Incorrect. DPO loss encourages preferred outputs by maximizing their relative log-likelihood and discourages dispreferred outputs. **⑦ 방향 왜곡 (minimize↔maximize relative)**
- (d) Incorrect. No simulation or multi-agent system is involved. **허구 메커니즘**

참고 수식: $\mathcal{L}_{\mathrm{DPO}} = -\log\sigma\left(\beta\log\frac{\pi_\theta(y_w|x)}{\pi_{\mathrm{ref}}(y_w|x)} - \beta\log\frac{\pi_\theta(y_l|x)}{\pi_{\mathrm{ref}}(y_l|x)}\right)$ — chosen·rejected의 implicit reward **차이를 키우는**(상대 log-likelihood 최대화) 방향.

4-4. Pick all that are correct · 4 pts

2025 중간 기출

기말 연관: #12 RL (DPO 한계 → SimPO/GRPO 동기)

Which of the following are limitations of DPO highlighted in recent studies?

a. DPO requires a separate reward model trained on human preferences.

b. DPO causes computational overhead due to the reference model.

c. DPO tends to generate shorter outputs regardless of quality.

d. DPO is inherently an online learning algorithm.

▼ 정답 & 해설 보기

정답: (b) — 단독 정답

- (a) Incorrect. Unlike RLHF, DPO does not require a separate reward model.
  ④ 방법 귀속 (RLHF의 특징)
- (b) Correct. DPO still depends on a reference model, which adds compute cost.
- (c) Incorrect. Studies show DPO can induce **longer** outputs unintentionally.
  방향 반전 (shorter↔longer)
- (d) Incorrect. DPO is based on fixed preference datasets (offline), unlike online RLHF.
  ⑥ 범주 혼동 (online↔offline)

4-5. Pick all that are correct · 4 pts

2025 중간 기출

기말 연관: #8 AI Feedback (evaluation bias·LC win rate) — 직결!

Which of the following are true about evaluation bias introduced by DPO?

a. Longer responses often receive higher preference scores from human evaluators.

b. The length-controlled (LC) win rate metric considers differences in instruction difficulty.

c. Ignoring length bias in evaluation can lead to misleading alignment conclusions.

d. LC win rate is a type of perplexity-based evaluation.

▼ 정답 & 해설 보기

정답: (b), (c)

- (a) Incorrect. Bias toward longer outputs has been observed in **LLM** evaluations. **① 주체 바꿔치기 (human⇔LLM evaluator)**
- (b) Correct. LC win rate uses regression to capture length and difficulty effects.
- (c) Correct. Evaluating without controlling for length may misrepresent performance.
- (d) Incorrect. LC win rate is preference-based, not perplexity-based. **⑥ 범주 혼동**

주의: (a)는 가장 까다로운 함정 — "길이 bias" 자체는 사실이지만 출제자는 "human evaluators"를 짚어 Incorrect 처리했다(강의에서는 LLM judge의 length bias로 다룸). 주체를 끝까지 읽어라.

6 Problem 5: Decoding (15 Points) 2025 중간 기출 원문

"Answer the following questions about alignment of LLMs." (← 원문의 오타: 실제 주제는 Decoding) — 모든 문항: Pick all that are correct; any wrong answer will lead to 0 points.

기말 연관 최고 Problem 5 전체가 기말 범위 **#7 Decoding/Controllability**와 직결 — 중간·기말에 걸친 챕터이므로 **기말에서 변형 재출제 가능성이 가장 높은 5문항**이다.

5-1. Pick all that are correct · 3 pts 2025 중간 기출 #7 Decoding

Which of the following are true about autoregressive decoding in LLMs?

a. Each token is generated by conditioning on all previous tokens.
b. The decoding process can be parallelized across all tokens during inference.
c. The process continues until an end-of-sequence (EOS) token is generated or upon reaching a pre-defined max length.
d. The next token is sampled from a fixed Gaussian distribution.

▼ 정답 & 해설 보기

정답: (a), (c)

- (a) Correct. Standard autoregressive decoding generates one token at a time, using all the previous tokens.
- (b) Incorrect. It is inherently sequential (not parallel). ⓔ training(병렬)↔inference(순차) 혼동
- (c) Correct. Generation typically stops when an EOS token is generated or predefined threshold is reached.
- (d) Incorrect. Tokens are sampled from the model's output probability distribution, not Gaussian. 허구 메커니즘

5-2. Pick all that are correct · 3 pts 2025 중간 기출 #7 Decoding

Which of the following are true about temperature-controlled sampling?

a. Temperature controls the sharpness or smoothness of the output distribution.

b. Lower temperature values result in more deterministic outputs.

c. Temperature is irrelevant in top-k sampling.

d. Temperature is applied after the softmax operation.

▼ 정답 & 해설 보기

정답: (a), (b)

- (a) Correct. High temperature → more randomness, low temperature → peaky distribution.
- (b) Correct. Lower temperatures lead to more confident predictions.
- (c) Incorrect. Temperature is still relevant and often used alongside top-k. ② 절대어 ("irrelevant")
- (d) Incorrect. It rescales **logits before** applying softmax. ⑤ 처리 순서 함정

참고 수식: $p_i = \frac{\exp(z_i/T)}{\sum_j \exp(z_j/T)}$ — temperature T는 softmax **이전** logits z를 나눈다.

5-3. Pick all that are correct · 3 pts 2025 중간 기출 #7 Decoding

Which of the following statements are correct about top-p (nucleus) sampling?

a. Top-p sampling adjusts the candidate set based on cumulative probability.

b. Top-p is a special case of beam search with p=1.

c. Top-p allows dynamic flexibility based on the distribution's shape.

d. It is designed to avoid unnatural or degenerate completions.

▼ 정답 & 해설 보기

정답: (a), (c), (d)

- (a) Correct. The candidate pool is chosen based on cumulative probability and predefined threshold.
- (b) Incorrect. Top-p is not related to beam search. ④ 방법 귀속 (sampling↔search)
- (c) Correct. It adapts the number of candidates per step.

- (d) Correct. It was designed to mitigate degeneration in sampling.

5-4. Pick all that are correct · 3 pts

2025 중간 기출

#7 Decoding (speculative decoding)

Which of the following describe speculative decoding?

a. It uses a small draft model to propose multiple tokens at once.

b. It verifies the drafts with a larger target model.

c. It eliminates the need for sequential generation.

d. It speeds up decoding by reducing the number of expensive large model inferences.

▼ 정답 & 해설 보기

정답: (a), (b), (d)

- (a) Correct. A lightweight model proposes token drafts.
- (b) Correct. The large target model checks and filters the drafts.
- (c) Incorrect. The draft model still generates the tokens via autoregressive generation.
  ② 절대어 ("eliminates")
- (d) Correct. It reduces the number of calls to the large model.

5-5. Pick all that are correct · 3 pts

2025 중간 기출

#7 Decoding (test-time compute)

Which of the following describe the concept of test-time compute scaling in LLMs?

a. It allows models to dynamically allocate more inference compute to harder queries.

b. It is a technique for fine-tuning large language models using reinforcement learning.

c. It enables more accurate reasoning by allowing "longer thinking" on complex tasks.

d. It improves model capacity by increasing the number of parameters at test time.

▼ 정답 & 해설 보기

정답: (a), (c)

- (a) Correct. Dynamic inference strategies help allocate more resources to harder problems.

- (b) Incorrect. Test-time compute scaling is not a fine-tuning method, but an inference strategy. **⑥ 범주 혼동 (training↔inference)**
- (c) Correct. It enables the model to produce better results by spending more compute.
- (d) Incorrect. The number of parameters is fixed; it increases inference time, not model size. **⑥ compute↔parameters 혼동**

7 기말 범위 출제 예측 (#7~#14)

7.1 예상 시험 구성 마지막 수업 공지 반영

마지막 수업에서 교수가 직접 공지: **객관식 Problem은 3~4개이고, 그중 1개는 #13 Jailbreak + #14 AI Text Detection 통합 문제**(*"I will make three or four multiple choice question problems, and one of them will be about the jailbreaking and AI-generated detection, all in one part"*). 중간 패턴(T/F 30점 + Pick-all 4개 Problem)을 여기에 결합하면 아래 구성이 가장 유력하다.

예상 Problem	예상 주제 (묶음)	예상 배점	근거
Problem 1	**True/False 10문항** — #7~#14 전 챕터에서 1~2개씩	30점	추정 교수는 "객관식 3~4개"만 언급 — 중간 패턴상 T/F는 별도 Problem로 유지될 가능성 큼 (무응답 1점 규칙 포함)
객관식 ①	**#9 RAG** — retriever 기초 + inference-level vs training-level	15~20점	추정 강의 3회분으로 분량 최대 + 교수 recap에서 이분법(아래 7.2) 명시적 강조
객관식 ②	**#10 Tool Use + #11 LLM Agents**	15~20점	추정 recap에서 "agents의 tool use는 이전 강의에서 다룸" — 연속 주제로 한 Problem 묶음이 자연스러움
객관식 ③	**#12 RL with LLMs** (+#8 AI Feedback 연계)	15~20점	추정 중간 Problem 4(Alignment)의 직계 후속 — recap에서 RLHF + PPO·GRPO 명시
객관식 ④	**#13 Jailbreak + #14 AI Text Detection 통합** (한 Problem 안에 두 주제 혼합)	15점	확정 교수 직접 공지 — 두 특강(녹화 강의)을 "all in one part"로 출제. 객관식이 3개면 ①~③ 중 둘이 합쳐질 수 있음

⚠️ 남은 불확실성

- **객관식이 3개일 경우** — 특강 통합 1개는 고정이므로, 본 범위(#7~#12)가 2개 Problem로 압축된다(예: RAG+Tool Use / Agents+RL). 어느 쪽이든 특강 Problem은 확정.

- #7 Decoding은 중간(Problem 5)에서 이미 출제됐고 **교수의 마지막 recap에서도 별도 언급 없음** — T/F로 축소되거나 다른 Problem에 흡수될 가능성이 높다. 단, 누적 범위라면 Problem 5 변형 재출제도 대비.
- T/F Problem의 존재·배점은 추정 — 시험 당일 표지의 채점 규칙(무응답 1점 여부)을 반드시 먼저 확인.

7.2 챕터별 "여기서 나온다" 포인트

아래 각 챕터의 recap 언급 표시는 **마지막 수업에서 교수가 범위 recap으로 직접 짚은 항목** — 출제 신호로 우선 암기.

#7 Decoding / Controllability 기출 직격: Problem 5 + P1(x) recap 미언급 — 우선순위 하향

- 기출에서 autoregressive·temperature·top-p·speculative·test-time compute가 **전부** 나옴 → 기말은 남은 조각: **greedy vs beam vs DBS 비교, top-k vs top-p 차이, contrastive/constrained decoding, controllability 기법**.
- T/F 단골: "Beam search *always* finds the global optimum" → False. "Temperature $T \to 0$ reduces sampling to greedy decoding" → True.
- 함정 재료: temperature 적용 위치(⑤), speculative의 draft↔target 역할 교환(①), "eliminates sequential generation"(②).

#8 AI Feedback 기출 직격: P4-5 (LC win rate) recap 언급

- recap 언급 교수가 짚은 키워드: **G-Eval**(평가) · **Anthropic HHH**(학습 개념) · **RLAIF = Constitutional AI** — 이 셋의 역할 구분(평가용 vs 정렬용)이 출제 신호.
- LLM-as-a-Judge의 bias 목록(**length/verbosity bias, position bias, self-enhancement bias**)은 P4-5의 직접 연장 — 각 bias의 정의와 완화법(swap positions, LC regression)이 Pick-all로 나오기 좋다.
- RLAIF vs RLHF: "무엇이 human을 대체하는가"(preference labeling 단계) — 주체 바꿔치기(①) 함정 1순위.
- Constitutional AI: critique→revision은 **SL 단계**, AI 비교는 **RL 단계** — 단계 귀속(④⑥) 함정.

#9 RAG 분량 최대 — 단독 Problem 유력 recap 언급

- recap 언급 교수가 recap에서 강조한 구조: retriever 기초 → **inference-level(REPLUG, IRCoT, RetRobust) vs training-level(Self-RAG, Search-R1)** 이분법. 이 분류 자체가

⑥(training↔inference) 함정 제조기 — **"REPLUG는 training-level이다" → False** 류의 귀속 판정 1순위.

- Sparse vs Dense retrieval: **BM25 vs DPR vs Contriever** 비교(학습 필요 여부, lexical↔semantic matching) — 방법 귀속(④) 함정 공장.
- "RAG *always* improves factuality / *eliminates* hallucination" → False(②). "Retrieval happens after generation" → False(⑤).
- RAG 파이프라인 단계(retriever→reader), end-to-end 학습 여부(RAG/REALM vs frozen retriever), retrieval 시점(when-to-retrieve, Self-RAG 류)이 Pick-all 소재.
- 수치 함정 후보: DPR negative sampling 방식, in-batch negatives, top-k passages 수.

#10 Tool Use recap 언급

- recap 언급 교수의 recap 구조: prompting 기초 + training-level 방법 2개 — **Toolformer = self-supervised, ToolLLM = SFT(supervised fine-tuning)**. 이 한 줄 구분이 recap에서 명시된 만큼 ④(방법 귀속) 함정의 최우선 재료.
- **Toolformer vs ToolLLM**: 누가 self-supervised로 API call을 삽입·필터링하나(Toolformer), 누가 대규모 실제 API + DFSDT인가(ToolLLM) — ④ 함정 1순위.
- Toolformer 필터링 기준: API call이 **future token loss를 줄이는가** — 목적 왜곡(⑦) 주의.
- "Tool use requires fine-tuning the LLM" → 모델별로 다름(prompting-only도 있음) — 절대어(②) 주의.

#11 LLM Agents recap 언급

- recap 언급 교수: agent = planning&action + tool use + memory인데 **"tool use는 이전 강의에서 다뤘으므로 두 개념 중심"** — 즉 **planning&action(ReAct, Reflexion, ToT)**과 **memory(MemGPT, A-Mem)**이 출제 축. MemGPT(OS식 메모리 계층) vs A-Mem(동적 메모리 조직)도 귀속 판정 후보.
- **ReAct vs Reflexion vs Tree-of-Thoughts** 3자 비교표가 출제 1순위: interleaved reasoning+acting(ReAct), verbal self-reflection을 episodic memory에 저장(Reflexion, *파라미터 업데이트 없음!*), search over thoughts + backtracking(ToT).
- "Reflexion updates model weights through reflection" → **False** (⑥ training↔in-context) — 가장 나올 법한 T/F.
- Agent 구성요소(planning, memory, tool use, action)와 multi-agent 협업 패턴.

#12 RL with LLMs 기출 직격: P4-2~4 + P1(ix) recap 언급

- recap 언급 교수 recap: **RLHF + 최신 알고리즘 PPO·GRPO** — 두 알고리즘이 recap에서 이름으로 호명된 유이한 RL 방법.
- **PPO vs GRPO**: GRPO는 **value/critic model 제거**, group 내 상대 보상으로 advantage 추정 $\hat{A}_i = \frac{r_i - \mathrm{mean}(r)}{\mathrm{std}(r)}$ — "GRPO requires a critic" → False(④).
- RLVR(verifiable rewards): rule-based reward(정답 일치·형식)로 reward model 불필요 — "RLVR trains a reward model" → False.
- online(PPO/GRPO) vs offline(DPO/SimPO) 구분(⑥)은 기출 P4-4(d)의 재탕이 유력.
- KL penalty의 역할(reference model에서 멀어지는 것 방지)과 적용 위치(⑤).

#13 Jailbreak (Special Lecture) 통합 Problem 확정 — #14와 한 문제

- recap 언급 교수가 recap에서 호명한 3축: **Jailbroken**(NeurIPS'23, jailbreak 개념을 정식화한 **베이스라인**) · 공격 **AutoDAN**(jailbreak prompt 자동 생성) · 방어 **SafeDecoding + Gradient Cuff**. 이 4개 이름의 **공격/방어 귀속 판정**이 통합 Problem의 핵심 재료.
- 우선순위 힌트 recap에서 **GCG·COLD-Attack은 언급되지 않음**(AutoDAN·Jailbroken 중심) — 출제된다면 보조 보기 수준일 가능성. 단 Jailbroken의 실패 원인 2축(competing objectives, mismatched generalization)은 베이스라인이므로 필수.
- 함정 재료: "AutoDAN is a defense method" → False(④), "SafeDecoding modifies the attack prompt" → False(방어는 디코딩 단계 개입 — ⑤), white-box↔black-box 혼동(①).

#14 AI Text Detection (Special Lecture) 통합 Problem 확정 — #13과 한 문제

- recap 언급 교수가 recap에서 호명한 계보: **DetectGPT → Fast-DetectGPT**(계산 시간 단축) **→ ReMoDetect**(다른 방식의 구별) + **watermarking**(생성 텍스트에 숨은 패턴 삽입, 회사 측만 검출 가능 — "이미 실제 배포됨").
- 교수 재강조 직관 *"AI 생성 텍스트는 모든 토큰에서 높은 확률을 받지만, 인간 텍스트는 모델 분포에서 나온 게 아니므로 확률이 출렁인다(fluctuate)"* — 이 직관을 묻는 보기(T/F든 Pick-all이든)가 나올 확률 최상.
- 함정 재료: "Watermarking can be applied to text after it is generated" → False(생성 중 개입 — ⑤), "Fast-DetectGPT improves accuracy, not speed" → False(동기 왜곡 ⑦), "Anyone can detect the watermark by reading the text" → False(회사 측 검출).

7.3 특강 통합형 예상 문제 — Pick all that are correct 한 Problem에 #13+#14 혼합 확정

교수 공지대로 **jailbreak와 detection 보기가 한 문제 안에 섞여 나올 수 있는** 통합형 대비 문제. 기출 문체(각 보기 = 독립 T/F)로 작성.

Pred-1 · Pick all that are correct 예상 문제 (#13+#14 통합)

Pick all the correct statements about LLM safety attacks and defenses, and AI-generated text detection.

- (a) AutoDAN is an attack method whose goal is to automatically craft jailbreak prompts.
- (b) SafeDecoding defends against jailbreak attacks by intervening at the decoding stage rather than by retraining the model from scratch.
- (c) DetectGPT requires training a separate supervised classifier on labeled human/AI text pairs.
- (d) Watermarking detects AI-generated text by analyzing the probability curvature of the text after it is generated.

▼ 정답 & 해설 보기

정답: (a), (b)

- (a) **Correct.** AutoDAN automatically generates stealthy jailbreak prompts (attack side).
- (b) **Correct.** SafeDecoding is a decoding-time defense — it contrasts token probabilities to amplify safety-aligned behavior, with no full retraining.
- (c) **Incorrect.** DetectGPT is zero-shot — it uses probability curvature under perturbations, not a trained classifier. ④ 방법 귀속
- (d) **Incorrect.** Watermarking intervenes *during generation* (hidden pattern in token selection); probability curvature is DetectGPT's mechanism. ④ 귀속 + ⑤ 시점

Pred-2 · Pick all that are correct 예상 문제 (#13+#14 통합)

Pick all the correct statements regarding jailbreaking and AI-generated text detection methods covered in the special lectures.

- (a) The work "Jailbroken" (NeurIPS 2023) attributes jailbreak success to two failure modes of safety training: competing objectives and mismatched generalization.
- (b) Gradient Cuff is an attack method that uses gradient information to optimize adversarial suffixes.
- (c) AI-generated text tends to receive consistently high probabilities across all tokens under the generating model, whereas human-written text shows fluctuating probabilities.

- (d) Fast-DetectGPT improves over DetectGPT mainly by reducing computational cost.

▼ 정답 & 해설 보기

정답: (a), (c), (d)

- (a) **Correct.** Jailbroken is the baseline work formalizing why safety training fails (competing objectives / mismatched generalization).
- (b) **Incorrect.** Gradient Cuff is a *defense* (detects jailbreak prompts via the refusal-loss landscape); gradient-optimized adversarial suffixes describe GCG.
  ① attacker⇔defender + ④ 귀속
- (c) **Correct.** 교수가 마지막 수업에서 재강조한 핵심 직관 — decoding methods sample high-probability tokens, so AI text is uniformly high-probability while human text fluctuates.
- (d) **Correct.** DetectGPT's perturbation-based curvature estimation is expensive; Fast-DetectGPT's main contribution is efficiency.

Pred-3 · Pick all that are correct

예상 문제 (#13+#14 통합)

Pick all the correct statements about watermarking and detection-based approaches for identifying AI-generated text.

- (a) Watermarking imposes hidden patterns during text generation that are invisible to ordinary readers but detectable by the model provider.
- (b) Watermarking can be applied retroactively to any text after it has been generated.
- (c) DetectGPT identifies AI-generated text in a zero-shot manner by measuring how the log-probability drops when the text is perturbed.
- (d) ReMoDetect follows exactly the same probability-curvature mechanism as DetectGPT, only with a faster implementation.

▼ 정답 & 해설 보기

정답: (a), (c)

- (a) **Correct.** 교수 recap 그대로 — hidden pattern은 사람 눈에는 안 보이고, 회사(provider) 측 검출 방법으로만 확인 가능. 이미 실제 배포됨.
- (b) **Incorrect.** Watermarking must intervene *during* generation (biasing token selection); it cannot be added after the fact. ⑤ 시점

- (c) **Correct.** AI text sits at a local maximum of the model's log-probability (negative curvature) — perturbation drops its probability sharply; zero-shot, no classifier training.
- (d) **Incorrect.** Faster implementation of the same curvature idea is *Fast-DetectGPT*; ReMoDetect takes a different route (reward-model-based detection exploiting LLMs' alignment preference). ④ 귀속

Pred-4 · Pick all that are correct 예상 문제 (#13+#14 통합)

Pick all the correct attributions of the following methods to their roles (attack / defense / detection).

- (a) AutoDAN — attack; SafeDecoding — defense; DetectGPT — detection.
- (b) Jailbreaking attacks succeed only against open-source models whose gradients are accessible.
- (c) Both SafeDecoding and Gradient Cuff are defense methods against jailbreak attacks.
- (d) Detection methods such as DetectGPT and watermarking both require access to the model side (the generating model's probabilities or the provider's key) rather than working from the raw text alone.

▼ 정답 & 해설 보기

정답: (a), (c), (d)

- (a) **Correct.** 정확한 3분류 — recap에서 교수가 호명한 그대로(공격 AutoDAN / 방어 SafeDecoding·Gradient Cuff / 탐지 DetectGPT 계열).
- (b) **Incorrect.** "only"가 함정 — Jailbroken/AutoDAN 류 prompt-level 공격은 black-box(API-only) 모델에도 통한다. gradient 접근이 필요한 것은 GCG 류뿐. ② 절대어
- (c) **Correct.** 둘 다 방어 — SafeDecoding은 decoding 개입, Gradient Cuff는 refusal loss 기반 jailbreak prompt 탐지.
- (d) **Correct.** DetectGPT needs the model's token probabilities; watermark detection needs the provider's secret method — neither works from surface text inspection by an arbitrary reader.

⭐ **최종 우선순위 (시간 없을 때) — 마지막 수업 공지 반영**

1. **#13+#14 특강 통합 Problem** 출제 확정 — 유일하게 출제가 확정된 Problem. 공격/방어/탐지 귀속표(AutoDAN↔SafeDecoding·Gradient Cuff, DetectGPT 계보, watermarking 시점) + "AI 텍스트는 전 토큰 고확률, 인간 텍스트는 출렁임" 직관. 위 7.3 예상 문제 4개 풀기.
2. **#12 RL(RLHF + PPO vs GRPO) + #8(G-Eval·HHH·RLAIF, judge bias)** — 중간 Problem 4의 직계 후속 + recap 호명.
3. **#9 RAG(inference-level REPLUG·IRCoT·RetRobust vs training-level Self-RAG·Search-R1)** — recap에서 이분법 명시 + 분량 최대.
4. **#10+#11(Toolformer self-supervised vs ToolLLM SFT, ReAct/Reflexion/ToT + MemGPT/A-Mem)** — recap 호명 키워드.
5. **기출 Problem 5 재독(#7)** — recap 미언급으로 우선순위는 내려갔지만 T/F 1~2문항 대비.

🎯 3분 요약

항목	한 줄 정리
형식	T/F 10×3점(무응답 1점!) + Pick-all 20문항(부분점수 0) = 100점/100분
T/F 전략	50:50이면 찍기(기대값 1.5 > 1), 완전히 모르면 비워서 1점 확보
Pick-all 전략	보기 4개를 독립 T/F로 판정 — 정답 수는 1~4개 전부 가능
함정 7유형	①주체 바꿔치기 ②절대어 ③수치 ④방법 귀속 ⑤처리 순서 ⑥training/inference·online/offline 범주 ⑦목적 왜곡
확정 공지	객관식 Problem 3~4개, 그중 1개 = **#13 Jailbreak + #14 Detection 통합** (마지막 수업 교수 발언). 기말 30%·절대평가·A+ 엄격
기출 직결 챕터	#7(Problem 5 전체+DBS), #8(P4-5 LC win rate), #12(P4-2~4 RLHF/DPO+SimPO)
기말 1순위	특강 통합(공격/방어/탐지 귀속 + 토큰 확률 직관), RL(PPO vs GRPO), RAG(inference vs training-level), Agents(ReAct vs Reflexion vs ToT + MemGPT/A-Mem)

← 전체 목차

전체 목차로 →